Arya Sarkar

Senior AI Engineer | LLM • RAG • Backend

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Summary

Senior AI Engineer with 4+ years of experience building production AI systems that combine LLMs, retrieval, tool use, and agent orchestration to solve real-world problems. Led the development of multi-agent products including conversational AI platforms, document intelligence systems, and AI-powered creative tools. Strong background in Python, backend architecture, evaluation, observability, and scaling AI applications from prototype to production.

Work Experience

Founding AI Engineer & Backend Lead 2021/03 – Present
Docunexus Inc. Remote
• Built Kognie Research, a multi-agent document intelligence platform leveraging RAG, retrieval pipelines, vector search, and LLM reasoning to enable conversational search, analysis, and question-answering across enterprise knowledge bases.
• Architected backend services, APIs, asynchronous workflows, AI orchestration layers, and observability systems supporting production AI applications across multiple products.
• Led a remote, cross-functional team of 5 engineers and designers to design, build, and launch AI-powered products from 0→1.
• Built Kognie Graphic Studio, a production multi-agent AI design platform where specialized agents collaborate on creative planning, prompt generation, image generation, design critique, iterative refinement, and editing workflows.
• Designed agent orchestration, evaluation pipelines, tool-use workflows, and model routing systems across multiple foundation models to improve output quality, reliability, and cost efficiency.
• Implemented monitoring, debugging, tracing, and observability systems for distributed AI services, agent workflows, and model-serving infrastructure.
• Collaborated directly with customers and stakeholders to translate business requirements into production-ready AI capabilities.

AI Research Fellow 2025/06 – 2025/08
Picampus Remote
- Developed a low-latency speech-to-speech translation system delivering human-comparable translation quality with 3–4 second streaming latency.
- Built and optimized real-time inference pipelines supporting technical lectures, scientific content, and terminology-aware translation workflows.

Summer Research Fellow (OSRE) 2024/04 – 2024/09
OSPO, UC Santa Cruz Remote
- Worked on reproducibility in interactive data systems
- Contributed to system design improvements for data workflows and tooling

Managing Fellow 2023/02 – 2023/10
Engineering for Change, LLC Remote
- Managed a team of 5 researchers to facilitate research and development w.r.t UNs SDGs.
- Published report in open-access journals, and helped advance workflow by creating conversational AI applications in the fintech and education domains.

Research Fellow 2022/05 – 2022/09
Engineering for Change, LLC Remote
Worked with a non-profit, JFF Labs to analyze the state of data maturity, and data literacy in the USA to generate an all inclusive research work following UN's SDG 4 and 8.

Summer Research Fellow 2021/05 – 2021/08
MITACS Remote
Worked at the TEEL Lab, University of Waterloo, Canada for a 3 month-long fully funded research program in the field of Ophthalmic Diagnosis using DL.

Education

B.Tech - Computer Science Engineering 2018/07 – 2022/05
University of Engineering and Management, Kolkata Kolkata, India

AISSCE (10+2) 2016 – 2018
B.D.M.International Kolkata, India
Finished my higher-secondary in the Computer Science stream.

AISSE (10th Grade) 2015 – 2016
B.D.M.International Kolkata, India

Skills

AI & LLMs ●●●●○
Large Language Models (LLMs), RAG, Multi-Agent Systems, LangChain, LangGraph, Prompt Engineering, AI Evaluation,

Vector Search & Databases ●●●●○
Pinecone, Vector Databases, PostgreSQL, NoSQL

Infrastructure ●●●○○
Docker, Linux, Git, CI/CD, AWS, GCP, Observability

Backend & APIs ●●●●○
Python, FastAPI, Flask, REST APIs, Async Processing, Laravel

Data & ML ●●●●○
Scikit-Learn, Pandas, NumPy, Machine Learning, Deep Learning

Projects

Kognie Graphic Design Studio
AI Image Studio
- Built and deployed a production-ready multi-agent AI design platform for content generation, image editing, and design automation.
- Led a team of 4 developers and designers to take the platform from 0->1

Kognie Research Engine
Multi Agentic RAG System for Document Intelligence
Designed and developed a RAG-powered document intelligence platform using OpenAI, LangChain, and Pinecone, enabling accurate conversational search, retrieval, and question answering across enterprise knowledge bases.

ELT Near Real-Time Translation System
Conversational AI assistant
- Built a production-ready speech-to-speech translation platform delivering human-comparable quality with 3–4 second latency.
- Optimized for educational and technical content with specialized terminology enhancement and mathematical content detection.

Publications

Imbalance aware Multimodal Attention Enabled Latent Space Oversampling for Emotion Recognition 2024/03
Springer

COVID-19 chest X-ray detection through blending ensemble of CNN snapshots 2022/07
Elsevier

3D Human action Recognition: Through the Researchers Eyes 2022/01
Elsevier

Automated Detection and Diagnosis of Diabetic Retinopathy: A Comprehensive Survey 2021/08
MDPI

Awards

Best Research Paper Award 2024/03
CIPR, 6th Internation Conference
Received the best research paper award for my submission to CIPR-2024, the 6th International Conference.

Best Research Paper Award 2020/06
OPTRONIX, 6th International Conference
Received the best research paper award for my submission to OPTRONIX, the 6th International Conference

Best Startup Award 2019/04
UEM, Kolkata, 2019
Received the chancellor's award for the best startup venture out of many startups.